UNIVERSITY OF ESWATINI

CSC 411: INTEGRATIVE PROGRAMMING TECHNOLOGIES

PART 1: REQUIREMENTS ANALYSIS OF A JUSTICE ADMINISTRATION SYSTEM

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1. Introduction

Justice administration is a process in which several institutions cooperate to move a suspect from arrest, through court proceedings, to custody or release. The CSC 411 mini-project brief identifies the **Police**, the **Judiciary** and the **Correctional Services** as the organisations involved. Today these organisations exchange paper files and phone calls, which causes delays, lost records and inconsistent information. The shared goal of the proposed system is that case information and authorised decisions move between the three institutions in a timely, traceable and secure way.

This document describes the process, identifies the stakeholders and their roles, states the initial functional and non-functional requirements and presents a Data Flow Diagram (DFD). The design is a proposal for academic purposes.

In Eswatini the organisations are the **Royal Eswatini Police Service**, the **Judiciary of Eswatini** (Magistrates Courts and the High Court, with the Director of Public Prosecutions presenting cases) and the **Correctional Services**. Bail and trial procedure are mainly governed by the **Criminal Procedure and Evidence Act 67 of 1938** (as amended) and the **Constitution of the Kingdom of Eswatini Act 2005**.

2. Description of the Inter-Organisational Process

Three organisations, each with its own information system and database, cooperate through REST interfaces. A unique reference number is used for the same matter in each system: the Police case ID (CASE-), the court case ID (CRT-) and the inmate ID (INM-), and the suspect is identified across systems by the national ID. Each organisation sees only the data it is authorised to access.

2.1 Main activities

1. **Suspect registration and arrest.** A complainant reports an alleged offence to the Police. A police officer records the report on the complainant's behalf, registers the suspect (name, national ID, date of birth, address) and records the arrest (date, place, arresting officer).
2. **Case creation and charging.** The Police create a case with a unique reference and add one or more charges (offence and statute) to it. Information is checked for completeness before the case leaves the Police.
3. **Referral to the Judiciary.** The Police refer the case, with the suspect and charge details, to the Judiciary through a secure interface. The Judiciary verifies the information, registers a court case and returns its court case ID.

4. **Court proceedings.** Court staff schedule hearings and decide whether the suspect is granted **bail** or held on **remand**. If the suspect is remanded, the Judiciary asks the Correctional Services to admit the suspect into custody.
5. **Verdict and sentence.** The court records the verdict. If the suspect is found guilty and sentenced to imprisonment, the sentence is sent to the Correctional Services; if not guilty, the case is closed and the court orders the release of any remanded suspect. The Judiciary sends this release order to the Correctional Services, which release the inmate.
6. **Custody management and release.** Correctional officers record the admission, view the inmate status, transfer the inmate to another facility and release the inmate when the sentence is completed or the court orders it.
7. **Status updates and record keeping.** Every authorised update is recorded with a timestamp, the responsible user and the case reference, so the progress of a case can be traced from arrest to release.

3. Stakeholders and Their Roles

Stakeholder	Role in the process	Information / responsibility	System role
Police (Royal Eswatini Police Service)	Receive reports, arrest and investigate.	Register suspects, record arrests, create cases, add charges and refer cases.	police_officer
Judiciary / court staff (Magistrates Courts, High Court)	Administer court proceedings.	Receive referrals; schedule hearings; record bail, remand, verdicts and sentences.	court_officer
Correctional Services	Hold and manage persons in custody.	Admit inmates; update status; transfer and release inmates.	correctional_officer
Complainants / reporting persons	Provide the initial information.	Report incidents to the Police. They are external stakeholders and do not use the system directly.	none
Prosecutors and legal representatives	Support the legal process.	Use authorised case materials according to court procedures (outside the scope of this demo).	none
System administrators / ICT support	Keep the system available and secure.	Manage user accounts, monitor system health, read audit logs, without unnecessary access to case content.	admin

The roles of prosecutors and legal representatives are included as supporting stakeholders only; their exact access depends on agreement with the institutions and applicable law.

4. Initial Requirements Analysis

4.1 Functional requirements

- The system shall authenticate users and give each user a role (police officer, court officer, correctional officer or administrator).
- The system shall allow police officers to register a suspect, record an arrest and create a case with a unique reference.
- The system shall allow charges to be added to an open case.
- The system shall support secure referral of a case, with the suspect and charge details, from the Police to the Judiciary.
- The system shall allow the Judiciary to schedule hearings and to record bail, remand, verdict and sentence.

- The system shall support secure transmission of remand and sentence information of admission, sentence and release orders from the Judiciary to the Correctional Services.
- The system shall allow the Correctional Services to admit, transfer and release inmates and to view an inmate's status.
- The system shall allow authorised users to search for and view the latest status of a case or inmate using its reference.
- The system shall record an audit trail of important actions with the time, user and record involved.

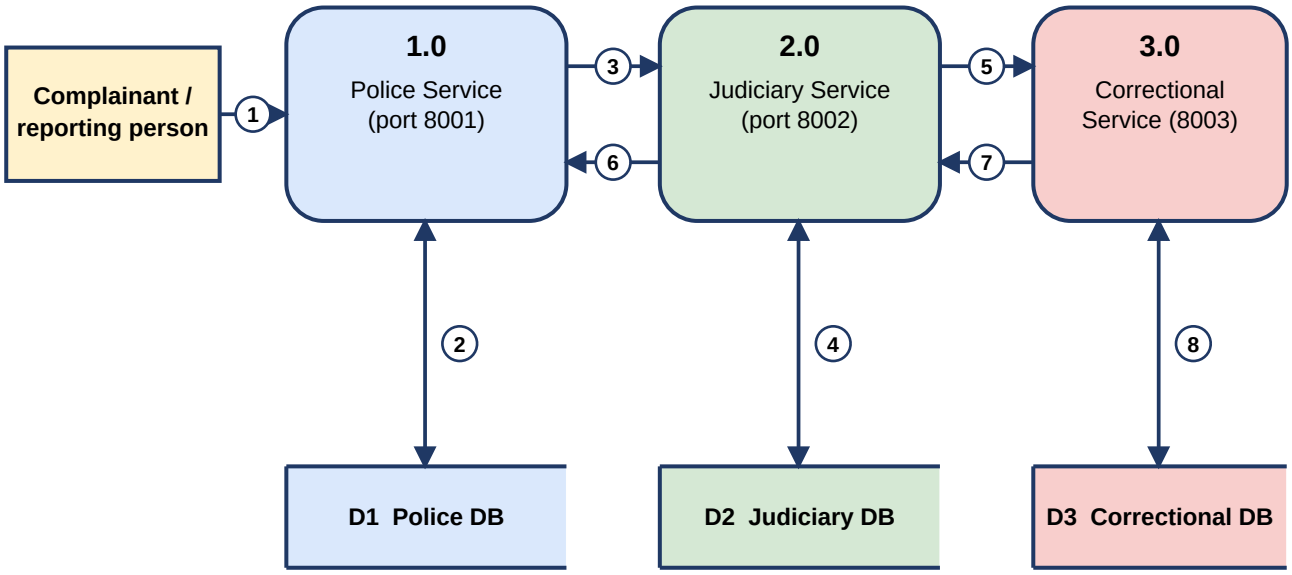
4.2 Non-functional requirements

- **Security and confidentiality.** Sensitive information must be protected through authentication (JWT), role-based access and secure communication.
- **Data integrity.** The system should reduce duplicate records and prevent unauthorised or accidental changes; invalid status changes are rejected.
- **Availability and reliability.** Authorised staff should be able to use the system when required, with backup and recovery arrangements; if one service is unavailable, the others report a clear error.
- **Performance.** Common actions such as searching for a case or retrieving its status should complete within about two seconds under normal workloads.
- **Interoperability.** The institutional systems exchange agreed data through documented REST interfaces (OpenAPI) in JSON format.
- **Usability.** Screens and workflows should be understandable to trained staff and give clear validation messages.
- **Auditability.** Important reads, updates, referrals and status changes should be logged so that authorised reviewers can trace activity.
- **Maintainability.** Each organisation's service is modular, so that a change to one component can be made with limited disruption to the others.

5. Proposed Data Flow Diagram (DFD)

Figure 1 shows the process at a conceptual level. The complainant reports to the Police (flow 1). The Police service stores its own records (2) and refers the case to the Judiciary (3). The Judiciary stores court records (4), and when a suspect is remanded or sentenced it requests admission at the Correctional Services, and when the court orders a release (not guilty or bail) it sends a release order (5). Confirmations return to the sender (6 and 7), and the Correctional Services keep their own inmate records (8). There is **no shared database**: each organisation owns its data store and the systems exchange information only through their REST interfaces.

Level 0 DFD: Inter-organisational Justice Administration System



Numbers refer to the data flows in the table below. Each service has its own data store.

Figure 1: Level 0 DFD of the Justice Administration System.

No.	Data flow	From	To
1	Incident report and suspect details	Complainant	1.0 Police Service
2	Suspect, arrest, case and charge records (read/write)	1.0 Police Service	D1 Police DB
3	Case referral (suspect and charges)	1.0 Police Service	2.0 Judiciary Service
4	Court case, hearings, bail, remand and verdict records (read/write)	2.0 Judiciary Service	D2 Judiciary DB
5	Admission request (remand or sentence) or release order	2.0 Judiciary Service	3.0 Correctional Service
6	Referral confirmation with court case ID	2.0 Judiciary Service	1.0 Police Service
7	Admission confirmation with inmate ID	3.0 Correctional Service	2.0 Judiciary Service
8	Inmate, transfer and release records (read/write)	3.0 Correctional Service	D3 Correctional DB

This is a high-level conceptual diagram and not a verified description of existing government systems.

6. Conclusion

The proposed Justice Administration System connects the Police, the Judiciary and the Correctional Services around a shared case reference and controlled information exchange. The main activities are suspect registration and arrest, case creation and charging, referral to court, court proceedings with bail or remand, verdict and sentence, and custody management and release. The requirements emphasise secure access, reliable records, interoperability, audit trails and timely status updates. This analysis is the foundation for Part 2, which designs the integrated REST-based architecture using Python, FastAPI and SQLite, with one database per service.

References

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